Chapter 1. Linear Regression and Gradient Descent

Lectures: Week 1 Wed (Regression, normal equations, gradient descent, generalization)

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1 The Regression Problem

In regression we observe pairs $(\mathbf{x}_i, y_i)$, $i = 1, \dots, n$, with input $\mathbf{x}_i \in \mathbb{R}^k$ and real response y_i, and we want a function that predicts y from $\mathbf{x}$. The simplest choice is a linear function. The lecture motivates the choice of "best line" by the sum of squared errors (SSE): a horizontal benchmark line gives one SSE, rotating it changes the SSE, and plotting SSE against the rotation angle gives a curve that first decreases and then increases. The best line is the minimiser of this curve.

Definition: linear regression model and design matrix

$$y_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + \varepsilon_i, \qquad i = 1, \dots, n.$$

Absorb the intercept with a constant input 1: let $d = k + 1$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top \in \mathbb{R}^d$, and let the **design matrix** $X \in \mathbb{R}^{n \times d}$ have ith row $(1, x_{i1}, \dots, x_{ik})$. Then

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad f(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 = (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

Rows of X index samples; columns index features.

2 Matrix Calculus Toolkit

We use the gradient convention: the gradient of a scalar with respect to a vector is a column vector of the same size.

Theorem: basic gradients

For $\mathbf{a} \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$,

$$\nabla_{\boldsymbol{\beta}} (\mathbf{a}^\top \boldsymbol{\beta}) = \mathbf{a}, \qquad \nabla_{\boldsymbol{\beta}} (\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = (A + A^\top)\boldsymbol{\beta}, \qquad \nabla^2_{\boldsymbol{\beta}} (\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = A + A^\top.$$

If A is symmetric the last two become $2A\boldsymbol{\beta}$ and $2A$.

Proof. $\mathbf{a}^\top \boldsymbol{\beta} = \sum_j a_j \beta_j$, so $\partial / \partial \beta_k = a_k$. For the quadratic form, $\boldsymbol{\beta}^\top A \boldsymbol{\beta} = \sum_{i,j} A_{ij} \beta_i \beta_j$. Differentiating with respect to β_k, the product rule picks up the terms with $i = k$ and those with $j = k$:

$$\frac{\partial}{\partial \beta_k} \sum_{i,j} A_{ij} \beta_i \beta_j = \sum_j A_{kj} \beta_j + \sum_i A_{ik} \beta_i = (A\boldsymbol{\beta})_k + (A^\top \boldsymbol{\beta})_k.$$

Differentiating once more gives the constant matrix $A + A^\top$. ■

3 Normal Equations

Theorem: least squares solution

$\nabla f(\boldsymbol{\beta}) = 2X^\top (X\boldsymbol{\beta} - \mathbf{y})$ and $\nabla^2 f = 2X^\top X \succeq 0$. Every minimiser solves the **normal equations**

$$X^\top X \boldsymbol{\beta} = X^\top \mathbf{y}.$$

If X has full column rank d, then $X^\top X \succ 0$ and the unique minimiser is $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$.

Proof. Expand $f = \mathbf{y}^\top\mathbf{y} - 2\mathbf{y}^\top X\boldsymbol{\beta} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta}$ (the two cross terms are equal scalars). With the toolkit, $\nabla f = -2X^\top\mathbf{y} + 2X^\top X\boldsymbol{\beta}$ and $\nabla^2 f = 2X^\top X$. For every $\mathbf{v}$, $\mathbf{v}^\top X^\top X\mathbf{v} = \|X\mathbf{v}\|^2 \geq 0$, so f is convex and stationary points are global minimisers. If the columns of X are linearly independent, $X\mathbf{v} \neq \mathbf{0}$ whenever $\mathbf{v} \neq \mathbf{0}$; then $X^\top X \succ 0$, it is invertible, f is strictly convex and the minimiser is unique. ■

Remark: geometry and rank

The normal equations say $X^\top(\mathbf{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}$: the residual is orthogonal to every column of X, so $X\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space. Since the first column is $\mathbf{1}_n$, the residuals sum to zero. Having $n > d$ does not by itself guarantee full rank; with dependent columns the minimiser is not unique and the pseudoinverse gives the minimum-norm solution. Numerically one solves the linear system by QR or SVD rather than forming an inverse.

Example: lecture example

Points $(0,1),(1,2),(2,2)$ and model $y = \beta_0 + \beta_1 x$:

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad X^\top\mathbf{y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad (X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}.$$

Hence $\hat{\boldsymbol{\beta}} = \frac{1}{6}(25-18, -15+18)^\top = (7/6,\ 1/2)^\top$. Predictions are $(7/6, 5/3, 13/6)$, residuals $(-1/6, 1/3, -1/6)$ (sum zero), and SSE $= 1/6$.

4 Gradient Descent

4.1 The algorithm

Instead of solving the normal equations directly we may iterate

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta\nabla f(\boldsymbol{\beta}_t) = \boldsymbol{\beta}_t + 2\eta X^\top(\mathbf{y} - X\boldsymbol{\beta}_t),$$

i.e. componentwise $\beta_j \leftarrow \beta_j + 2\eta\sum_i (y_i - \hat{y}_i)x_{ij}$ with $x_{i0} = 1$. The factor 2 is often absorbed into the learning rate η. Each full-batch step uses all samples. The negative gradient is the direction of steepest descent: for a unit vector $\mathbf{v}$ the first-order change $\nabla f^\top\mathbf{v}$ is minimised, by Cauchy–Schwarz, at $\mathbf{v} = -\nabla f/\|\nabla f\|$.

4.2 Convergence on least squares

Write $J(\boldsymbol{\beta}) = \frac{1}{2}\|X\boldsymbol{\beta} - \mathbf{y}\|^2$ (so $\nabla J = X^\top(X\boldsymbol{\beta} - \mathbf{y})$) and $H = X^\top X$, assumed positive definite.

Theorem: exact convergence condition

Gradient descent $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta\nabla J(\boldsymbol{\beta}_t)$ converges to $\hat{\boldsymbol{\beta}}$ for every initial point if and only if

$$0 < \eta < \frac{2}{\lambda_{\max}(X^\top X)}.$$

Proof. Error recursion. Since $\nabla J(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, $X^\top\mathbf{y} = H\hat{\boldsymbol{\beta}}$, so $\nabla J(\boldsymbol{\beta}_t) = H(\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}})$. With $\mathbf{e}_t = \boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}$,

$$\mathbf{e}_{t+1} = \mathbf{e}_t - \eta H\mathbf{e}_t = (\mathbf{I} - \eta H)\mathbf{e}_t, \qquad \mathbf{e}_t = (\mathbf{I} - \eta H)^t\mathbf{e}_0.$$

Diagonalise. H is symmetric, so $H = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$, $\lambda_i > 0$. Using $\mathbf{I} = QQ^\top$, $\mathbf{I} - \eta H = Q(\mathbf{I} - \eta\Lambda)Q^\top$. Let $\mathbf{c}_t = Q^\top \mathbf{e}_t$. Multiplying the recursion by $Q^\top$ and using $Q^\top Q = \mathbf{I}$ gives $\mathbf{c}_{t+1} = (\mathbf{I} - \eta\Lambda)\mathbf{c}_t$, a diagonal recursion:

$$c_{t,i} = (1 - \eta\lambda_i)^t \, c_{0,i}.$$

Because $\|\mathbf{c}_t\| = \|\mathbf{e}_t\|$, $\mathbf{e}_t \to \mathbf{0}$ iff every coordinate $c_{t,i} \to 0$.

Sufficiency. If $|1 - \eta\lambda_i| < 1$ for all i, every coordinate tends to zero for every $\mathbf{e}_0$.

Necessity. If $|1-\eta\lambda_i| \geq 1$ for some i, start from $\mathbf{e}_0 = \mathbf{q}_i$ (the ith eigenvector). Then $\mathbf{e}_t = (1-\eta\lambda_i)^t \mathbf{q}_i$ and $\|\mathbf{e}_t\| = |1 - \eta\lambda_i|^t \geq 1$, so this initial point does not converge.

Solve. $|1 - \eta\lambda_i| < 1 \iff 0 < \eta\lambda_i < 2 \iff 0 < \eta < 2/\lambda_i$. Requiring this for all i is equivalent to $0 < \eta < 2/\lambda_{\max}$. ■

Remark: conditioning

The contraction factor of direction i is $|1 - \eta\lambda_i|$. The rate $\max_i |1 - \eta\lambda_i|$ is minimised by $\eta^\star = 2/(\lambda_{\max} + \lambda_{\min})$, giving rate $(\kappa - 1)/(\kappa + 1)$ with condition number $\kappa = \lambda_{\max}/\lambda_{\min}$. Elongated (ill-conditioned) level sets make gradient descent slow; this is revisited in Chapter 8. At $\eta = 2/\lambda_{\max}$ the steepest direction oscillates with factor -1 and never decays, so the inequality is strict. In the lecture's example, $\lambda(X^\top X) = 4 \pm \sqrt{10}$ and the condition is $0 < \eta < (4 - \sqrt{10})/3 \approx 0.279$.

5 Polynomial Features and Generalization

5.1 Feature maps

For scalar input take $\boldsymbol{\phi}_M(x) = (1, x, x^2, \dots, x^M)^\top$ and $y(x) = \sum_{j=0}^{M} \beta_j x^j$. The model is nonlinear in x but linear in the coefficients, so all of the above applies with the (Vandermonde-type) design matrix whose rows are $\boldsymbol{\phi}_M(x_i)^\top$.

5.2 Training error versus generalization

Because a degree-M model contains every lower-degree model, the minimal training SSE cannot increase as M increases. Validation error can nevertheless increase: a high-degree fit follows noise and oscillates between data points (overfitting), while too low a degree cannot capture the trend (underfitting). Performance is measured by the root mean squared error

$$\mathrm{MSE} = \frac{1}{n}\mathrm{SSE}, \qquad E_{\mathrm{RMS}} = \sqrt{\mathrm{MSE}}.$$

Remark: roles of the data splits

Training data choose the coefficients; validation data choose the degree and the regularisation strength; a held-out test set estimates final performance. In the lecture's plots the gap between training and validation RMS grows with M for $M \geq 5$, and degree 9 with $n = 10$ overfits badly while $n = 100$ points support the same degree much better. "About ten points per parameter" is a heuristic, not a theorem. Regularisation (Chapter 4) is the second remedy.

6 Proof Supplement: Uniqueness and the Pseudoinverse

Theorem: set of minimisers

The minimisers of $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ are exactly the solutions of $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}$; this system is always consistent, and the solution set is $\hat{\boldsymbol{\beta}}_0 + \ker X$ for any particular solution $\hat{\boldsymbol{\beta}}_0$. Among them, $X^+\mathbf{y}$ has minimal Euclidean norm.

Proof. Consistency: $X^\top \mathbf{y}$ lies in the column space of $X^\top$, which equals the column space of $X^\top X$ (both have orthogonal complement $\ker X$, since $X^\top X\mathbf{v} = \mathbf{0} \Rightarrow \|X\mathbf{v}\|^2 = 0$). Two solutions differ by $\mathbf{v}$ with $X^\top X\mathbf{v} = \mathbf{0}$, i.e. $\mathbf{v} \in \ker X$, and give the same fitted values. Write $X = U\Sigma V^\top$ (thin SVD with r positive singular values). Every solution is $V_r\Sigma_r^{-1}U_r^\top \mathbf{y} + \mathbf{v}$ with $\mathbf{v} \perp \text{range}(V_r)$; by Pythagoras its norm is minimised at $\mathbf{v} = \mathbf{0}$, which is $X^+\mathbf{y}$. ■